

Data Analytics and Business Intelligence

Hands-on Exercise No. 3

DigiSkills 3.0 Batch-03

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My Solution File

1. Create calculated columns:

- To categorize sales into **High** and **Low** based on the Sales value, where values greater than 100 are labeled as High and others as Low. Name the new column as Sales Category.

Sales Category = IF([Sales] > 100, "High", "Low")													
te	City	Region	Postal Code	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit	Discount Category	Sales Category
formia	Los Angeles	West	90036	OFF-IA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters by Universal	14.62	2	0	6.8714	No Discount	Low
formia	Los Angeles	West	90032	FUR-FU-10001487	Furniture	Furnishings	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood	48.86	7	0	14.1694	No Discount	Low
formia	Los Angeles	West	90032	OFF-AR-10002833	Office Supplies	Art	Newell 322	7.28	4	0	1.9656	No Discount	Low
formia	Los Angeles	West	90032	OFF-AP-10002892	Office Supplies	Appliances	Belkin F5C206VTEL 6 Outlet Surge	114.9	5	0	34.47	No Discount	High
formia	San Francisco	West	94109	OFF-AR-10003056	Office Supplies	Art	Newell 341	8.56	2	0	2.4824	No Discount	Low
formia	Los Angeles	West	90049	TEC-AC-10003027	Technology	Accessories	Imation 8GB Mini TravelDrive USB 2.0 Flash Drive	90.57	3	0	11.7741	No Discount	Low
formia	Los Angeles	West	90049	OFF-ST-10003479	Office Supplies	Storage	Eldon Base for stackable storage shelf, platinum	77.88	2	0	3.894	No Discount	Low
formia	Los Angeles	West	90004	TEC-AC-10004633	Technology	Accessories	Verbatim 25 GB 6x Blu-ray Single Layer Recordable Disc, 3/Pack	13.98	2	0	6.1512	No Discount	Low
formia	Los Angeles	West	90004	OFF-PA-10003892	Office Supplies	Paper	Xerox 1943	146.73	3	0	68.9631	No Discount	High
formia	Los Angeles	West	90004	FUR-FU-10000397	Furniture	Furnishings	Luxo Economy Swing Arm Lamp	79.76	4	0	22.3328	No Discount	Low
formia	San Francisco	West	94122	OFF-AR-10002053	Office Supplies	Art	Premium Writing Pencils, Soft, #2 by Central Association for the Blind	14.9	5	0	4.172	No Discount	Low
formia	San Francisco	West	94122	OFF-ST-10002370	Office Supplies	Storage	Sortfilr Multipurpose Personal File Organizer, Black	21.39	1	0	6.2031	No Discount	Low
formia	Los Angeles	West	90036	OFF-AR-10004930	Office Supplies	Art	Turquoise Lead Holder with Pocket Clip	20.1	3	0	6.633	No Discount	Low
formia	Los Angeles	West	90036	OFF-PA-10000304	Office Supplies	Paper	Xerox 1995	6.48	1	0	3.1104	No Discount	Low
formia	Los Angeles	West	90004	FUR-FU-10003773	Furniture	Furnishings	Eldon Cleatmat Plus Chair Mats for High Pile Carpets	238.56	3	0	26.2416	No Discount	High
formia	Roseville	West	95661	OFF-PA-10001804	Office Supplies	Paper	Xerox 195	20.04	3	0	9.6192	No Discount	Low
formia	Roseville	West	95661	OFF-PA-10001736	Office Supplies	Paper	Xerox 1880	35.44	1	0	16.6568	No Discount	Low
formia	Roseville	West	95661	OFF-AR-10001149	Office Supplies	Art	Sanford Colorific Colored Pencils, 12/Box	11.52	4	0	3.456	No Discount	Low
formia	Roseville	West	95661	OFF-FA-10002988	Office Supplies	Fasteners	Ideal Clamps	4.02	2	0	1.9698	No Discount	Low
formia	Roseville	West	95661	OFF-SU-10001218	Office Supplies	Supplies	Friskars Softgrip Scissors	65.88	6	0	18.4464	No Discount	Low
formia	Roseville	West	95661	FUR-FU-10001706	Furniture	Furnishings	Longer-Life Soft White Bulbs	43.12	14	0	20.6976	No Discount	Low
formia	San Francisco	West	94122	OFF-AR-10000940	Office Supplies	Art	Newell 343	8.82	3	0	2.3814	No Discount	Low
formia	San Francisco	West	94122	OFF-EN-10004030	Office Supplies	Envelopes	Convenience Packs of Business Envelopes	10.86	3	0	5.1042	No Discount	Low
formia	San Francisco	West	94122	OFF-PA-10004327	Office Supplies	Paper	Xerox 1911	143.7	3	0	68.976	No Discount	High
formia	Pasadena	West	91104	OFF-ST-10003656	Office Supplies	Storage	Safco Industrial Wire Shelving	671.93	7	0	20.1579	No Discount	High
formia	San Jose	West	95123	OFF-PA-10001950	Office Supplies	Paper	Southworth 25% Cotton Antique Laid Paper & Envelopes	58.38	7	0	26.271	No Discount	Low
formia	San Jose	West	95123	OFF-PA-10002254	Office Supplies	Paper	Xerox 1883	105.52	4	0	48.5392	No Discount	High
formia	San Jose	West	95123	OFF-ST-10001590	Office Supplies	Storage	Tenex Personal Project File with Scoop Front Design, Black	80.88	6	0	21.0288	No Discount	Low
formia	Los Angeles	West	90045	OFF-PA-10002751	Office Supplies	Paper	Xerox 1920	5.98	1	0	2.691	No Discount	Low
formia	Los Angeles	West	90004	OFF-PA-10000659	Office Supplies	Paper	Adams Phone Message Book, Professional, 400 Message Capacity, 5 3/8" x 11"	20.94	3	0	9.8418	No Discount	Low
formia	Los Angeles	West	90004	OFF-PA-10001144	Office Supplies	Paper	Xerox 1913	110.96	2	0	53.2608	No Discount	High
formia	Los Angeles	West	90032	TEC-AC-10002323	Technology	Accessories	SanDisk Ultra 32 GB MicroSDHC Class 10 Memory Card	176.8	8	0	22.984	No Discount	High
formia	Redlands	West	92374	OFF-EN-10001509	Office Supplies	Envelopes	Poly String Tie Envelopes	14.28	7	0	6.7116	No Discount	Low
formia	Los Angeles	West	90032	OFF-ST-10001328	Office Supplies	Storage	Personal Filino Tote with Lid, Black/Gray	93.06	6	0	26.0568	No Discount	Low

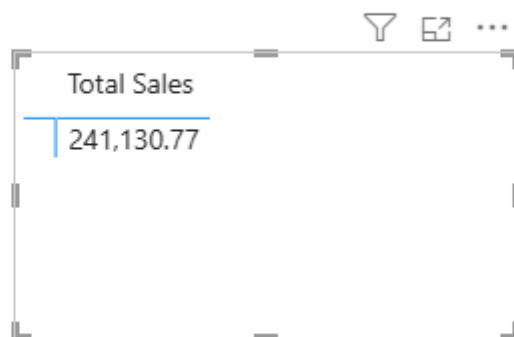
- To classify records based on discount, labeling entries with zero discount as “No Discount” and others as “Discount Applied”. Name the new column as Discount Category.

Discount Category = 2 If([Discount] = 0, "No Discount", "Discount Applied")														
Country	State	City	Region	Postal Code	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit	Discount Category	
United States	California	Los Angeles	West		90036	OFF-LA-100002340	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters by Universal	14.62	2	0	6.8714	No Discount
United States	California	Los Angeles	West		90032	FUR-FU-10001487	Furniture	Furnishings	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood	48.86	7	0	14.1694	No Discount
United States	California	Los Angeles	West		90032	OFF-AR-10002833	Office Supplies	Art	Newell 322	7.28	4	0	1.9656	No Discount
United States	California	Los Angeles	West		90032	OFF-AP-10002892	Office Supplies	Appliances	Belkin F5C206VTEL 6 Outlet Surge	114.9	5	0	34.47	No Discount
United States	California	San Francisco	West		94109	OFF-AR-10003056	Office Supplies	Art	Newell 341	8.56	2	0	2.4824	No Discount
United States	California	Los Angeles	West		90049	TEC-AC-10003027	Technology	Accessories	Imation 8GB Mini TravelDrive USB 2.0 Flash Drive	90.57	3	0	11.7741	No Discount
United States	California	Los Angeles	West		90049	OFF-ST-10003479	Office Supplies	Storage	Eldon Base for stackable storage shelf, platinum	77.88	2	0	3.894	No Discount
United States	California	Los Angeles	West		90004	TEC-AC-10004633	Technology	Accessories	Verbatim 25 GB 6x Blu-ray Single Layer Recordable Disc, 3/Pack	13.98	2	0	6.1512	No Discount
United States	California	Los Angeles	West		90004	OFF-PA-10003892	Office Supplies	Paper	Xerox 1943	146.73	3	0	68.9631	No Discount
United States	California	Los Angeles	West		90004	FUR-FU-10000397	Furniture	Furnishings	Luxo Economy Swing Arm Lamp	79.76	4	0	22.3328	No Discount
United States	California	San Francisco	West		94122	OFF-AR-10002053	Office Supplies	Art	Premium Writing Pencils, Soft, #2 by Central Association for the Blind	14.9	5	0	4.172	No Discount
United States	California	San Francisco	West		94122	OFF-ST-10002370	Office Supplies	Storage	Sortiflier Multipurpose Personal File Organizer, Black	21.39	1	0	6.2031	No Discount
United States	California	Los Angeles	West		90036	OFF-AR-10004930	Office Supplies	Art	Turquoise Lead Holder with Pocket Clip	20.1	3	0	6.633	No Discount
United States	California	Los Angeles	West		90036	OFF-PA-10000394	Office Supplies	Paper	Xerox 1995	6.48	1	0	3.1104	No Discount
United States	California	Los Angeles	West		90004	FUR-FU-10003773	Furniture	Furnishings	Eldon Cleatmat Plus Chair Mats for High Pile Carpets	238.56	3	0	26.2416	No Discount
United States	California	Roseville	West		95661	OFF-PA-10001804	Office Supplies	Paper	Xerox 195	20.04	3	0	9.6192	No Discount
United States	California	Roseville	West		95661	OFF-PA-10001736	Office Supplies	Paper	Xerox 1880	35.44	1	0	16.6568	No Discount
United States	California	Roseville	West		95661	OFF-AR-10001149	Office Supplies	Art	Sanford Colorific Colored Pencils, 12/Box	11.52	4	0	3.456	No Discount
United States	California	Roseville	West		95661	OFF-FA-10002988	Office Supplies	Fasteners	Ideal Clamps	4.02	2	0	1.9698	No Discount
United States	California	Roseville	West		95661	OFF-SU-10001218	Office Supplies	Supplies	Friskars Softgrip Scissors	65.88	6	0	18.4464	No Discount
United States	California	Roseville	West		95661	FUR-FU-10001706	Furniture	Furnishings	Longer-Life Soft White Bulbs	43.12	14	0	20.6976	No Discount
United States	California	San Francisco	West		94122	OFF-AR-10000940	Office Supplies	Art	Newell 343	8.82	3	0	2.3814	No Discount
United States	California	San Francisco	West		94122	OFF-EN-10004030	Office Supplies	Envelopes	Convenience Packs of Business Envelopes	10.86	3	0	5.1042	No Discount
United States	California	San Francisco	West		94122	OFF-PA-10004327	Office Supplies	Paper	Xerox 1911	143.7	3	0	68.978	No Discount
United States	California	Pasadena	West		91104	OFF-ST-10003656	Office Supplies	Storage	Safco Industrial Wire Shelving	671.93	7	0	20.1579	No Discount
United States	California	San Jose	West		95123	OFF-PA-10001950	Office Supplies	Paper	Southworth 25% Cotton Antique Laid Paper & Envelopes	58.38	7	0	26.271	No Discount
United States	California	San Jose	West		95123	OFF-PA-10002254	Office Supplies	Paper	Xerox 1883	105.52	4	0	48.5392	No Discount
United States	California	San Jose	West		95123	OFF-ST-10001590	Office Supplies	Storage	Tenex Personal Project File with Scoop Front Design, Black	80.88	6	0	21.0288	No Discount
United States	California	Los Angeles	West		90045	OFF-PA-10002751	Office Supplies	Paper	Xerox 1920	5.98	1	0	2.691	No Discount
United States	California	Los Angeles	West		90004	OFF-PA-10000659	Office Supplies	Paper	Adams Phone Message Book, Professional, 400 Message Capacity, 5 3/8" x 11"	20.94	3	0	9.8418	No Discount
United States	California	Los Angeles	West		90004	OFF-PA-10001144	Office Supplies	Paper	Xerox 1913	110.96	2	0	53.2608	No Discount
United States	California	Los Angeles	West		90032	TEC-AC-10002323	Technology	Accessories	SanDisk Ultra 32 GB MicroSDHC Class 10 Memory Card	176.8	8	0	22.984	No Discount
United States	California	Redlands	West		92374	OFF-EN-10001509	Office Supplies	Envelopes	Poly String Tie Envelopes	14.28	7	0	6.7116	No Discount
United States	California	Los Angeles	West		90032	OFF-ST-10001328	Office Supplies	Storage	Personal Filing Tote with Lid, Black/Gray	93.06	6	0	26.0568	No Discount
United States	California	Santa Clara	West		95051	OFF-PA-10003479	Office Supplies	Paper	Xerox 4200 Series MultiUse Premium Copy Paper (20Lb. and 84 Bright)	10.56	2	0	4.752	No Discount
United States	California	San Diego	West		92037	OFF-PA-10004569	Office Supplies	Paper	Wirebound Message Books, Two 4 1/4" x 5" Forms per Page	7.61	1	0	3.5767	No Discount

2. Create calculated measures:

- Total Sales (Sum of Sales)

1 Total Sales = SUM('final data'[Sales])



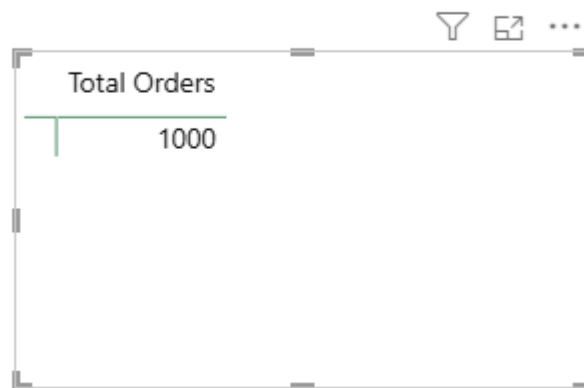
- Qty Sold (Sum of Quantity)

```
1 Qty Sold = SUM('final data'[Quantity])
```



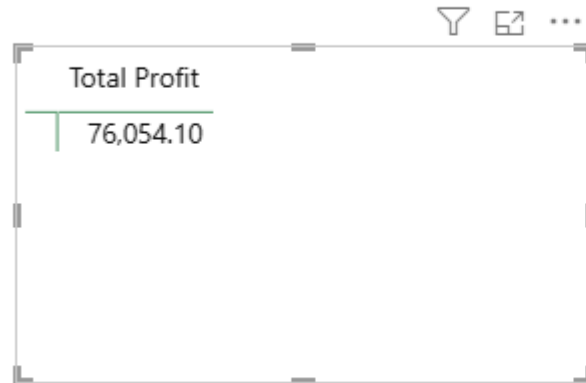
- Total Orders (Count of Order ID)

```
. Total Orders = COUNT('final data'[Order ID])
```



- Total Profit (Calculate by multiplying quantity by profit) using iterator function.

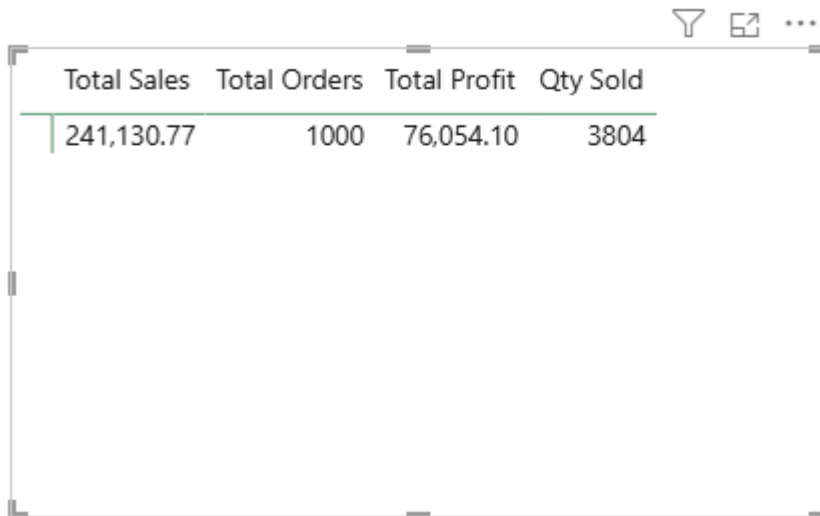
```
1 Total Profit = SUMX('final data', 'final data'[Quantity]* 'final data'[Profit])
```



A screenshot of a table visual showing a single column titled 'Total Profit'. The value '76,054.10' is displayed below the header. The table has a thin border and a small icon in the top right corner.

Total Profit
76,054.10

3. Validate all the results of calculated measures by using the Matrix visual.



A screenshot of a table visual showing four columns: 'Total Sales', 'Total Orders', 'Total Profit', and 'Qty Sold'. The values are '241,130.77', '1000', '76,054.10', and '3804' respectively. The table has a thin border and a small icon in the top right corner.

Total Sales	Total Orders	Total Profit	Qty Sold
241,130.77	1000	76,054.10	3804